THE GRAMMAR BEFORE THE FORMULA

IB Physics entry diagnostic • 25 minutes • 30 marks • no calculator

Name ______________________________ **Date** ______________

Instruction: do not reach for a formula. Decide what each sentence commits you to: what was observed, what was inferred, which quantities matter, and which relationship is being claimed.

1 | DESCRIPTION, PSEUDO-EXPLANATION OR EXPLANATION

6 MARKS • 6 MIN

Write **D** for a direct description or measurement, **P** for an explanation-shaped restatement/label, and **E** for a model-based explanation that links a process or interaction to evidence.

CODE	STATEMENT
☐	The water temperature fell from 78 °C to 64 °C in five minutes.
☐	The water cooled because cooling made its temperature go down.
☐	Energy was transferred from the warmer water to the cooler surroundings because of their temperature difference.
☐	When the magnet was moved closer, the paper clip began to move towards it.
☐	The clip moved because the magnet attracted it.
☐	A magnetic-field model predicts an interaction that exerts a force on the clip; changing separation changes the interaction.

2 | BUILD THE MODEL BEFORE THE EQUATION

8 MARKS • 8 MIN

Situation: A trolley is released from rest on a ramp. In successive 0.50 s intervals it travels 0.12 m, 0.31 m and 0.55 m. A steeper ramp produces larger increases.

Choose a system boundary. State whether Earth is inside or outside it. [1]

Name two quantities that were measured directly in the situation. [2]

State one relationship in words. Do not use symbols or a formula. [1]

Name one quantity or condition that should be held fixed when the ramp angle is changed. [1]

Underline the descriptive clause and box the explanatory clause: “The distance travelled in equal time intervals increased, because the component of the gravitational interaction along the ramp produced a resultant force.” [2]

What result would make you question that explanation? [1]

Checkpoint: a useful explanation risks being wrong. If no imaginable result could count against it, it is not yet doing much physics.

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3 | READ THE RELATIONSHIP INSIDE THE UNITS

8 MARKS • 5 MIN

Complete each relationship using words first, then give the compound unit. Spaces between unit symbols indicate multiplication; a negative exponent indicates division.

PHYSICAL IDEA	COMPLETE THE WORD RELATIONSHIP	UNIT
spring stiffness	force _________ extension	_________________
pressure	force _________ area	_________________
mechanical energy transfer	force acting _________ distance	_________________

PREFIX CHECK `200 g = ___________________ kg` `30 mm = ___________________ m`

4 | LANGUAGE FORENSICS

6 MARKS • 4 MIN

Match each sentence to one error family. Use each family once: **substance • agent • attribute • process-state • model-object**.

ERROR FAMILY	SENTENCE
_______________	A. The current is used up by the lamp.
_______________	B. Pressure pushes the cube upward.
_______________	C. The trolley carries a force with it.
_______________	D. The ball is accelerating, so it must be moving fast.
_______________	E. The straight line on the graph is the trolley’s path through the room.

Choose one sentence above and rewrite it so that the entities, interaction and quantity are represented more precisely. [1]

5 | EXIT WARRANT

2 MARKS • 2 MIN

A student says: **“The units work, so the explanation is correct.”**

What can a unit check legitimately tell you? [1]

Rewrite the claim so that it states the limitation of dimensional consistency. [1]